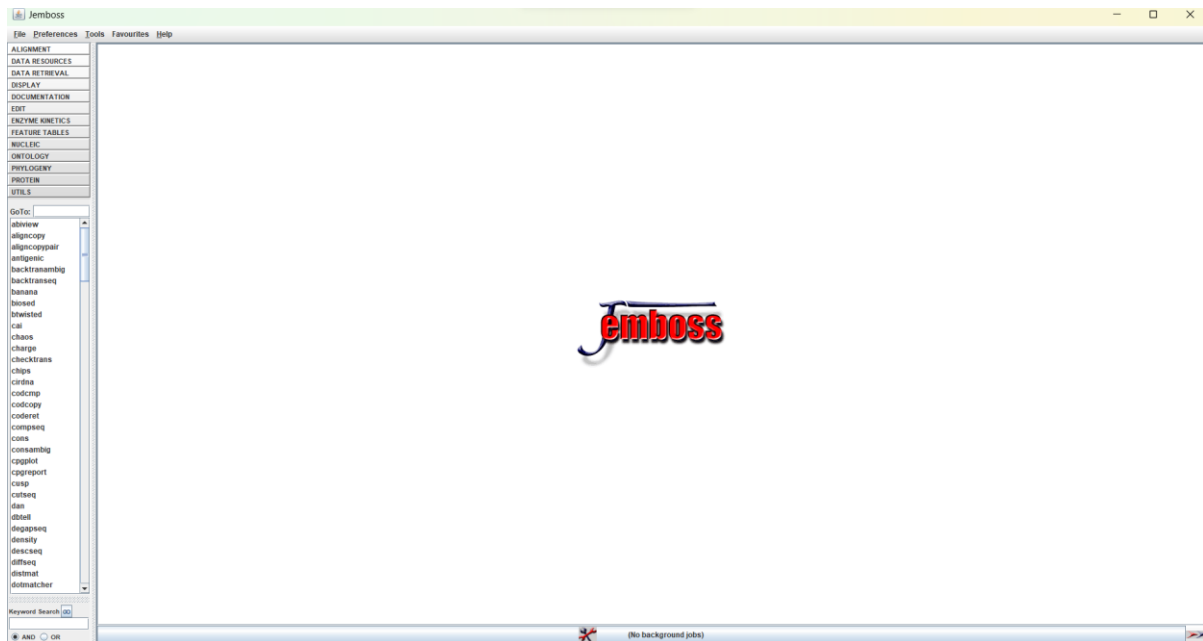




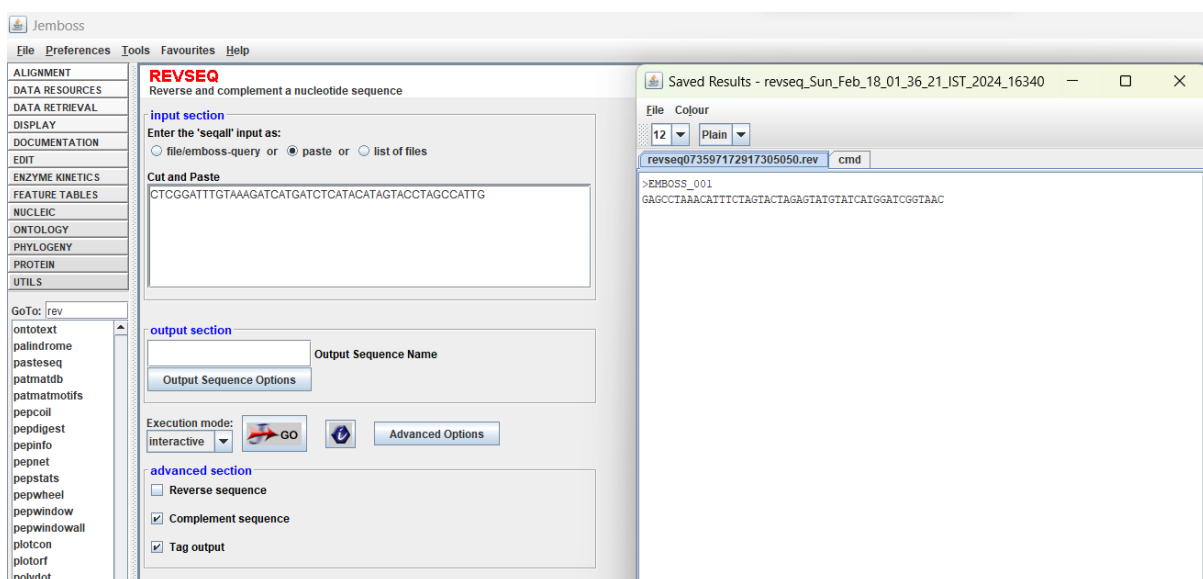
1. Successful installation of EMBOSS.



2. Using EMBOSS, find the complementary strand for the sequence:

5' - CTCGGATTGTGTAAGATCATGATCTCATACATAGTACCTAGCCATTG - 3'

3' - GAGCCTAAACATTTCTAGTACTAGAGTATGTATCATGGATCGGTAAC - 5'





3. A C++ program for finding the complementary strand for the sequence given in Q2.

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    string seq = "CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCATTG";
    //cin>>seq;    //Uncomment this line to take sequence input from user
    int n=seq.size();
    unordered_map<char,char> nucleotide_map = {
        {'A', 'T'}, {'T', 'A'}, {'G', 'C'}, {'C', 'G'},
    };
    string compSeq="";
    for(int i=0; i<n; i++)
    {
        compSeq+=nucleotide_map[seq[i]];
    }
    cout<<"Complementary sequence: ";
    cout<<compSeq<<"\n";
    return 0;
}
```

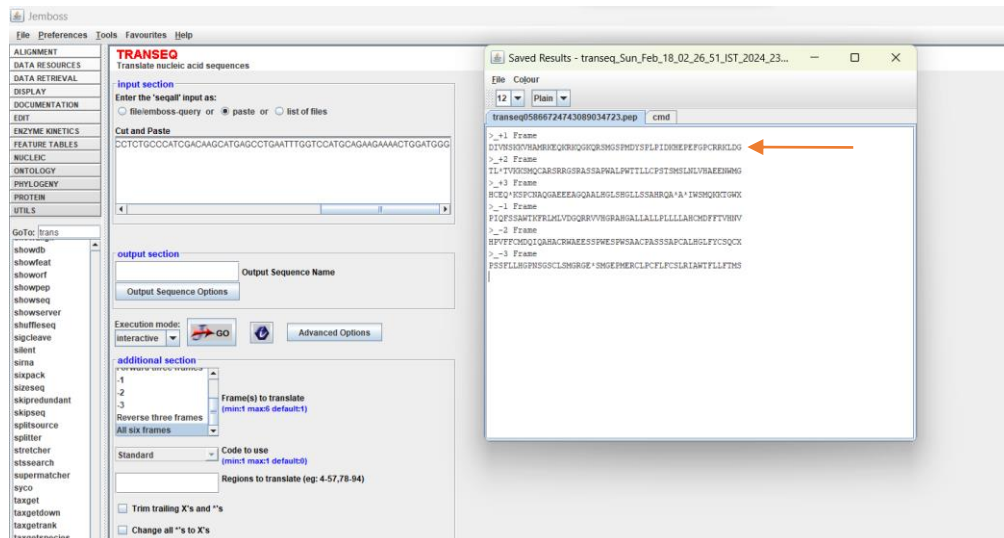
```
D:\programming\BT3040-A1\  ×  +  ▾
Complementary sequence: GAGCCTAAACATTTCTAGTACTAGAGTATGTATCATGGATCGGTAAC
Process returned 0 (0x0)   execution time : 0.090 s
Press any key to continue.
|
```



4.

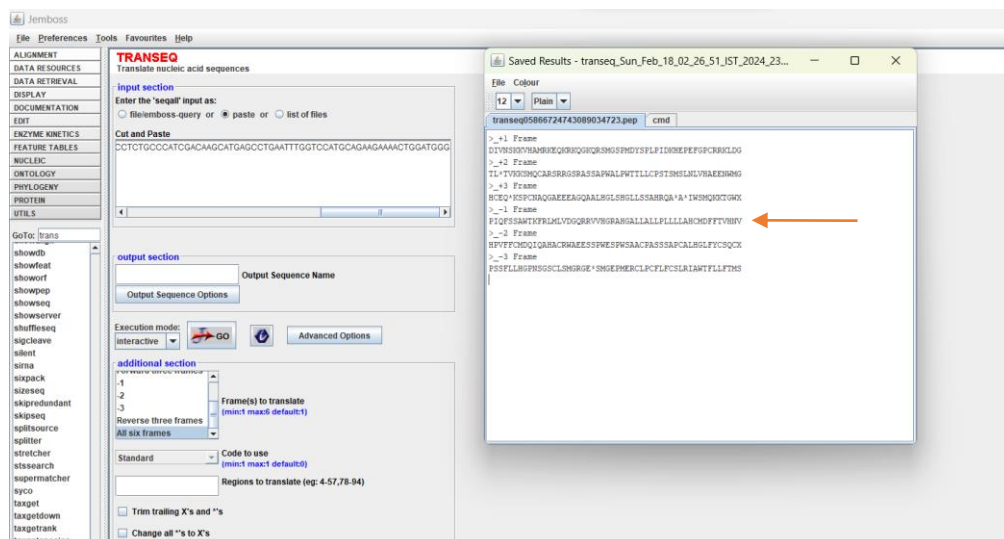
i. The translated sequence using EMBOSS:

DIVNSKKVHAMRKEQKRKQGGKQSRMSGSPMDYSPLPIDKHEPEFGPCRRKLDG



ii. The -1 reading frame matches with the given reading frame.

PIQFSSAWTKFRMLMLVDGQRRVHGRAHGALLALLPLLLLAHCMDFFTVHN





5. A C++ program to translate the given DNA sequence (from Q4).

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    string seq =
"GACATTGTGAACAGTAAAAAGTCCATGCAATGCGCAAGGAGCAGAAGAGGAAGCAGGGCAAGCAGCGCTCCAT
GGGCTCTCCCATGGACTACTCTCTCTGCCCATCGACAAGCATGAGCCTGAATTTGGTCCATGCAGAAGAAAAC
GGATGGG";

    //cin>>seq;    //Uncomment this line to take sequence input from user

    int n=seq.size();

    string protSeq="";

    unordered_map<string, char> codon_to_aa = {
        {"TTT", 'F'}, {"TTC", 'F'}, {"TTA", 'L'}, {"TTG", 'L'},
        {"CTT", 'L'}, {"CTC", 'L'}, {"CTA", 'L'}, {"CTG", 'L'},
        {"ATT", 'I'}, {"ATC", 'I'}, {"ATA", 'I'}, {"ATG", 'M'},
        {"GTT", 'V'}, {"GTC", 'V'}, {"GTA", 'V'}, {"GTG", 'V'},
        {"TCT", 'S'}, {"TCC", 'S'}, {"TCA", 'S'}, {"TCG", 'S'},
        {"CCT", 'P'}, {"CCC", 'P'}, {"CCA", 'P'}, {"CCG", 'P'},
        {"ACT", 'T'}, {"ACC", 'T'}, {"ACA", 'T'}, {"ACG", 'T'},
        {"GCT", 'A'}, {"GCC", 'A'}, {"GCA", 'A'}, {"GCG", 'A'},
        {"TAT", 'Y'}, {"TAC", 'Y'}, {"TAA", '*'}, {"TAG", '*'},
        {"CAT", 'H'}, {"CAC", 'H'}, {"CAA", 'Q'}, {"CAG", 'Q'},
        {"AAT", 'N'}, {"AAC", 'N'}, {"AAA", 'K'}, {"AAG", 'K'},
        {"GAT", 'D'}, {"GAC", 'D'}, {"GAA", 'E'}, {"GAG", 'E'},
        {"TGT", 'C'}, {"TGC", 'C'}, {"TGA", '*'}, {"TGG", 'W'},
        {"CGT", 'R'}, {"CGC", 'R'}, {"CGA", 'R'}, {"CGG", 'R'},
        {"AGT", 'S'}, {"AGC", 'S'}, {"AGA", 'R'}, {"AGG", 'R'},
        {"GGT", 'G'}, {"GGC", 'G'}, {"GGA", 'G'}, {"GGG", 'G'}
    };
};
```



```
for(int i=0; i<n; i+=3)
{
    if(i+2>=n)
        break;
    string str="";
    str+=seq[i]; str+=seq[i+1]; str+=seq[i+2];
    protSeq+=codon_to_aa[str];
}
cout<<"Protein sequence: ";
cout<<protSeq<<"\n";
return 0;
}
```

```
D:\programming\BT3040-A1\ x + v
Protein sequence: DIVNSKKVHAMRKEQKRKQGKQRSMGSPMDYSPLPIDKHEPEFGPCRRLDG
Process returned 0 (0x0)   execution time : 0.093 s
Press any key to continue.
|
```



6. A C++ code that searches for the given query subsequences in the main sequence and lists down the total number of matches and the start position of the match.

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    string seq =
"GACATTGTGAACAGTAAAAAAGTCCATGCAATGCGCAAGGAGCAGAAGAGGAAGCAGGGCAAGCAGCGCTCCAT
GGGCTCTCCCATGGACTACTCTCCTCTGCCCATCGACAAGCATGAGCCTGAATTTGGTCCATGCAGAAGAAACT
GGATGGG";

    int n=seq.size();

    map<string, int> seq_map_count = {
        {"AAG", 1}, {"GTC", 1}, {"GAG", 1}, {"ACTA", 1}, {"ATAT", 1},
    };

    map<string, vector<int>> seq_map_pos = {
        {"AAG", {}}, {"GTC", {}}, {"GAG", {}}, {"ACTA", {}}, {"ATAT", {}},
    };

    for(int i=0; i<n-2; i++)
    {
        string base=""; base+=seq[i]; base+=seq[i+1]; base+=seq[i+2];
        if(seq_map_count[base]>=1)
        {
            seq_map_count[base]++;
            seq_map_pos[base].push_back(i);
        }
    }

    for(int i=0; i<n-3; i++)
    {
        string base=""; base+=seq[i]; base+=seq[i+1]; base+=seq[i+2];
        base+=seq[i+3];
```



```
        if(seq_map_count[base]>=1)
        {
            seq_map_count[base]++;
            seq_map_pos[base].push_back(i);
        }
    }
for(auto &x:seq_map_count)
{
    if(x.second>=1)
    {
        cout<<"Enter the string: "<<x.first<<"\n";
        cout<<"Total Match: "<<x.second-1<<"\n";
        cout<<"Position of match: ";
        for(auto &y:seq_map_pos[x.first])
            cout<<y<<" ";
        cout<<"\n\n";
    }
}
return 0;
}
```

```
D:\programming\BT3040-A1\ >
Enter the string: AAG
Total Match: 7
Position of match: 19 36 45 51 60 111 140

Enter the string: ACTA
Total Match: 1
Position of match: 88

Enter the string: ATAT
Total Match: 0
Position of match:

Enter the string: GAG
Total Match: 3
Position of match: 39 47 117

Enter the string: GTC
Total Match: 2
Position of match: 21 130

Process returned 0 (0x0)   execution time : 0.121 s
Press any key to continue.
|
```

[illegible]

5' - CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCATTG - 3'

8



```
for(int i=0; i<n-1; i++)
{
    string base=""; base+=seq[i]; base+=seq[i+1];
    baseStackingEnergy+=base_stacking_energy_map[base];
}
cout<<"Average Base Stacking Energy: ";
cout<<baseStackingEnergy/(n-1)<<"\n";
return 0;
}
```

```
D:\programming\BT3040-A1\ x + v
Average Base Stacking Energy: -6.28261
Process returned 0 (0x0)   execution time : 0.013 s
Press any key to continue.
|
```



9. Computing the average melting temperature for the following sequences using Seq2Feature tool.

i. ATATATATAT

Your input seq is:
ATATATATAT

Physicochemical Properties:

Properties	Scale/unit	Average value
Stacking energy	kcal/mol	18
Enthalpy	kcal/mol	8.04444
Entropy	cal/mol/K	16.6222
Flexibility_shift	$\text{kJ mol}^{-1} \text{\AA}^{-2}$	2.53
Flexibility_slide	$\text{kJ mol}^{-1} \text{\AA}^{-2}$	9.66333
Free energy	kcal/mol	0.655556
Melting Temperature	degree	48.0022
Mobility to bend towards major groove	μ	1.09778
Mobility to bend towards minor groove	μ	1.03333
Probability contacting nucleosome core	%	6.75566
Rise stiffness	kcal/mol angstrom	7.80778
Roll stiffness	kcal/mol degree	19.3333
Shift stiffness	kcal/mol angstrom	0.892222
Slide stiffness	kcal/mol angstrom	2.66111
Tilt stiffness	kcal/mol degree	28
Twist stiffness	kcal/mol degree	25.8889

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ii. GCGCGCGCGC

Your input seq is:
GCGCGCGCGC

Physicochemical Properties:

Properties	Scale/unit	Average value
Stacking energy	kcal/mol	175556
Enthalpy	kcal/mol	110778
Entropy	cal/mol/K	27.5556
Flexibility_shift	$\text{kJ mol}^{-1} \text{\AA}^{-2}$	6.49111
Flexibility_slide	$\text{kJ mol}^{-1} \text{\AA}^{-2}$	4.19778
Free energy	kcal/mol	1.65889
Melting Temperature	degree	107.867
Mobility to bend towards major groove	μ	0.99778
Mobility to bend towards minor groove	μ	1.20556
Probability contacting nucleosome core	%	3.37778
Rise stiffness	kcal/mol angstrom	8.05333
Roll stiffness	kcal/mol degree	21.5556
Shift stiffness	kcal/mol angstrom	1.14667
Slide stiffness	kcal/mol angstrom	2.33889
Tilt stiffness	kcal/mol degree	31.5556
Twist stiffness	kcal/mol degree	20.1111

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Melting temperature for **sequence (i)** is found out to be **48.0022° C**, whereas the melting temperature for **sequence (ii)** is **107.867° C**. This almost more than double of a difference in their melting temperatures indicates high bonding energy between G and C compared to A and T and we know this is true as there is a triple hydrogen bond between G and C and a double hydrogen bond between A and T. Therefore, this huge difference in melting temperature can be explained with the above reasoning.



10. Calculating the AT and GC content of the sequence AAATGGCCCTAA using Seq2Feature tool.

Your input seq is:

AAATGGCCCTAA

Nucleotide Content:

	Nucleotide content in %
AT_content	58.333333
Adenine_content	41.666667
Cytosine_content	25.000000
GC_content	41.666667
Guanine_content	16.666667
Keto_GT_content	33.333333
Purine_AG_content	58.333333
Thymine_content	16.666667

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AT content: 58.33%

GC content: 41.67%

The End